

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2025

Course: B.Tech

Branch : Computer Engineering/Computer Science & Engineering

Semester : 6

Subject Code & Name: BTCOC603_Y23 Machine Learning

Max Marks: 60

Date:16/01/2026

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks				
Q. 1	Objective-type questions. (Compulsory Question)		12				
1	A model is trained to predict house prices from features like size, number of rooms, and location. The training data includes both the features and the actual sale prices. Which type of machine learning problem is this?		1				
	<table border="1"> <tr> <td>a. Regression</td> <td>b. Classification</td> <td>c. Supervised Learning</td> <td>d. Reinforcement Learning</td> </tr> </table>	a. Regression	b. Classification	c. Supervised Learning	d. Reinforcement Learning		
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2	You train a classifier and observe that it performs very well on the training data but poorly on unseen test data. Which situation does this most likely indicate?		1				
	<table border="1"> <tr> <td>a. Under fitting</td> <td>b. High Bias & Low Variance</td> <td>c. Good Generalisation</td> <td>d. Overfitting</td> </tr> </table>	a. Under fitting	b. High Bias & Low Variance	c. Good Generalisation	d. Overfitting		
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3	Which of the following is the main goal of using a validation set during model development?		1				
	<table border="1"> <tr> <td>a. To Provide Completely Unseen data for Final Reporting</td> <td>b. To Train the Model Parameters</td> <td>c. To Tune Hyperparameters & Select the Models</td> <td>d. To Increase the Size of The Training Set</td> </tr> </table>	a. To Provide Completely Unseen data for Final Reporting	b. To Train the Model Parameters	c. To Tune Hyperparameters & Select the Models	d. To Increase the Size of The Training Set		
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4	In a binary classification problem, which metric is most appropriate when the positive class is rare, and the cost of missing a positive case is very high?		1				
	<table border="1"> <tr> <td>a. Precision</td> <td>b. Accuracy</td> <td>c. Recall</td> <td>d. AUC ROC</td> </tr> </table>	a. Precision	b. Accuracy	c. Recall	d. AUC ROC		
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5	Which of the following best describes the bias-variance trade-off in machine learning models?		1				

	a. Bias is Always Harmful, Variance is Always Beneficial	b. Bias & Variance are Unrelated Concepts	c. Reducing Bias often increases variance and vice versa	d. Increasing Model Complexity always reduces both Bias & Variance		
6	What is the primary purpose of regularisation techniques such as L1 or L2 regularization in machine learning models?					1
	a.To Prevent Overfitting by Controlling Model Complexity	b. To Guarantee Perfect Accuracy on the Training Dataset	c. T Reduce Model Training Time to Zero	d. To Increase The Size of the Dataset		
7	Which algorithm is commonly used for both classification and regression problems					1
	a. k-Means	b. Decision Tree	c. PCA	d. Apriori		
8	What is the role of an activation function in a neural network?					1
	a.Reduce computational cost	b. Introduce non-linearity	c. Initialize weights	d. Normalize input data		
9	Which algorithm uses "Gradient Descent" for optimisation?					1
	a.K-Nearest Neighbours	b. Linear Regression	c.Naive Bayes	d.Random Forest		
10	In reinforcement learning, what does the agent learn from?					1
	a. Labelled data	b. Unlabelled data	c.Rewards and penalties	d.Feature correlations		
11	Which method is used for handling missing data?					1
	a. Deleting all incomplete rows	b. Replacing with random values	c. Imputation (e.g., mean, median)	d. Ignoring the issue		
12	Which evaluation metric is suitable for imbalanced classification?					1
	a. Accuracy	b. AUC-ROC	c. Mean Squared Error	d. R-squared		
Q. 2	Solve the following.					12
A)	Compare the bias-variance trade-off. How do they affect model performance?					6
B)	Explain how gradient descent optimises a model.					6

Q. 3	Solve the following.		12
A)	Describe the working of Random Forest. Why is it called "forest"?		6
B)	What is regularisation? Explain L1 (Lasso) vs. L2 (Ridge).		6
Q. 4	Solve Any Two of the following.		12
A)	Explain principal component analysis (PCA) and its applications.		6
B)	What are support vectors in SVM? How does kernel trick work?		6
C)	Explain backpropagation in neural networks with the chain rule.		6
Q. 5	Solve Any Two of the following.		12
A)	Explain ensemble learning with examples.		6
B)	What is the purpose of training, validation, and test datasets?		6
C)	How does K-Means clustering work?		6
Q. 6	Solve Any Two of the following.		12
A)	Explain how linear regression works, including the cost function and optimisation method.		6
B)	Explain SMOTE for handling imbalanced datasets.		6
C)	What is gradient descent? Differentiate batch, stochastic, and mini-batch variants.		6
	*** End ***		